Question ID: 6762772f

Question

Artificial leaves are a developing renewable energy technology that mimics the process of photosynthesis in plants. These devices are silicon-based solar cells coated in chemical catalysts that activate reactions that split water molecules into hydrogen and oxygen gas. The technology, while generating lots of interest, is not yet commercially viable as a large-scale energy source. To meet this challenge, scientists from many fields are researching ways to store, transport, and distribute the energy the devices produce while other scientists are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

Answer

A. Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.

B. The recent increase in the commercial use of artificial leaves as an energy source has encouraged many scientists to research ways to improve the technology.

C. Artificial leaves split water molecules into oxygen and hydrogen gas using catalysts more efficiently than plants do using the process of photosynthesis.

D. Artificial leaves were developed to mimic the natural process of photosynthesis in plants in order to store energy for long-term commercial use.